

Wearable Alert Device (WAD): Critical Design Review

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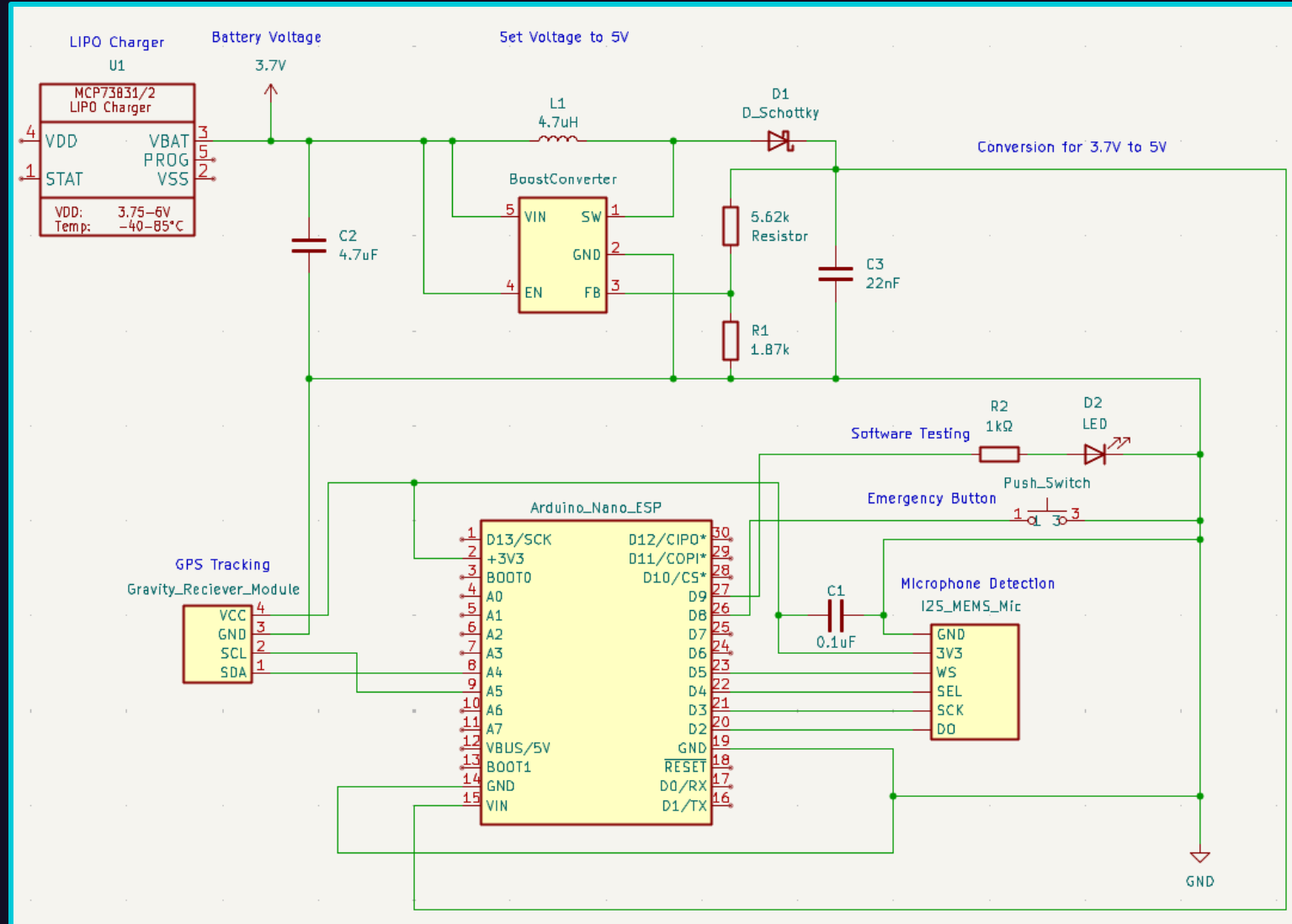
► Requirements

- Complete build on protoboard
- Software implementation and completion
- Full testing of individual features
- Future proofing and how we can improve

► Schedule

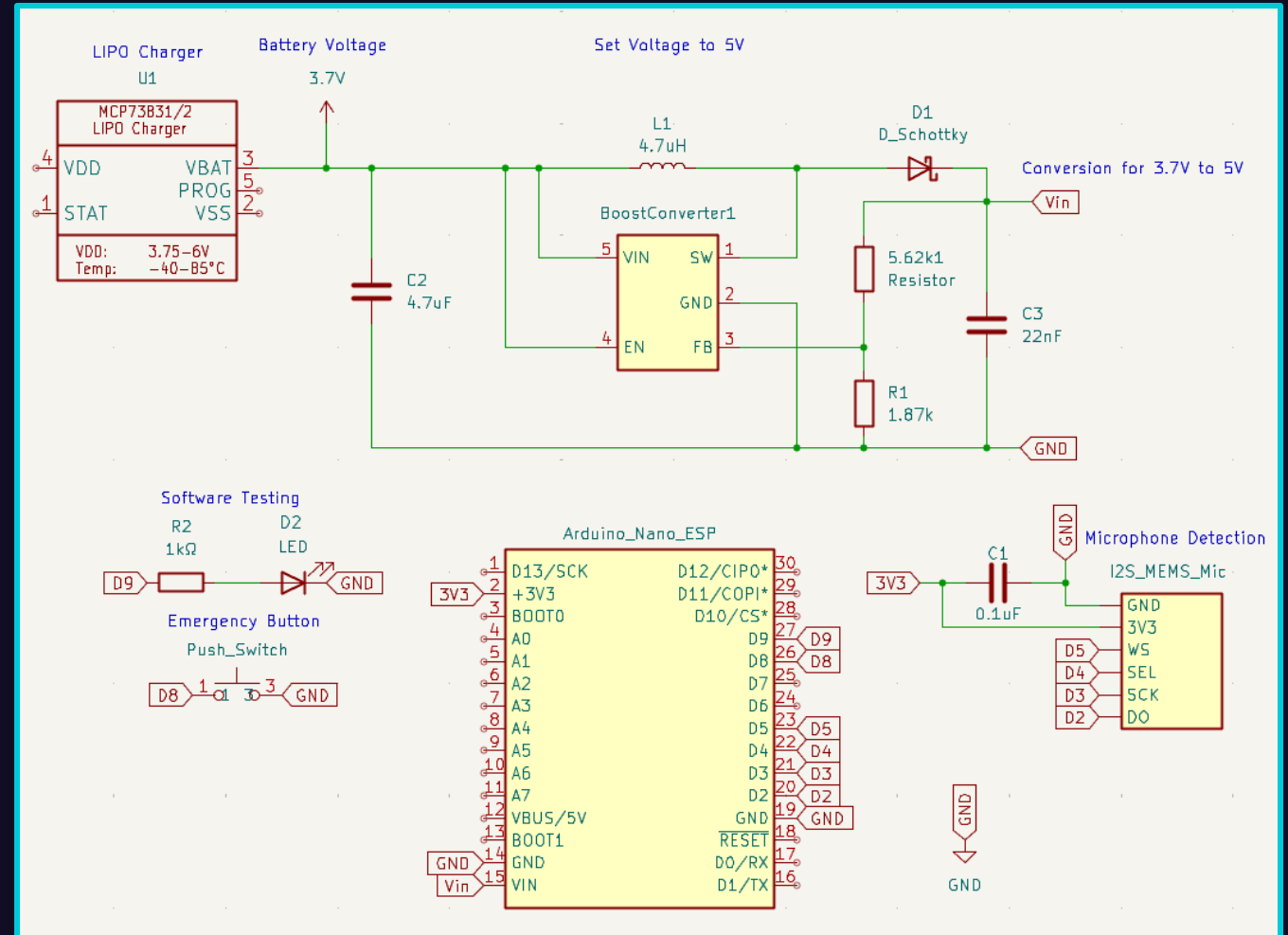
	Mar 22nd	Mar 29th	Apr 5th	Apr 12th	Apr 19th	Apr 26th
Parts Arrival						
Protoboard Building						
Testing						
Final Demo						

Schematics Overview

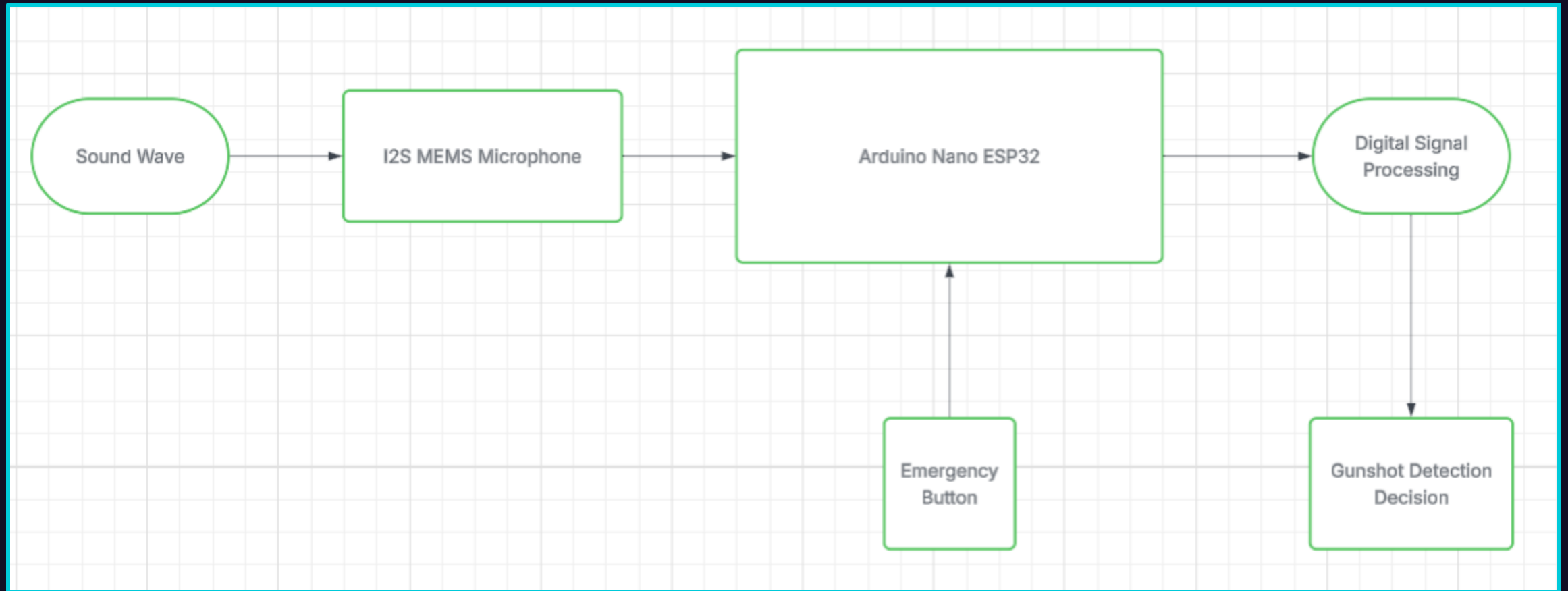


Gunshot Detection Schematics

- I2S MEMS Microphone
 - Operation Voltage: 3V3
 - Decoupling Capacitor
 - Digital Connections
- Emergency Button
- LED Software Testing

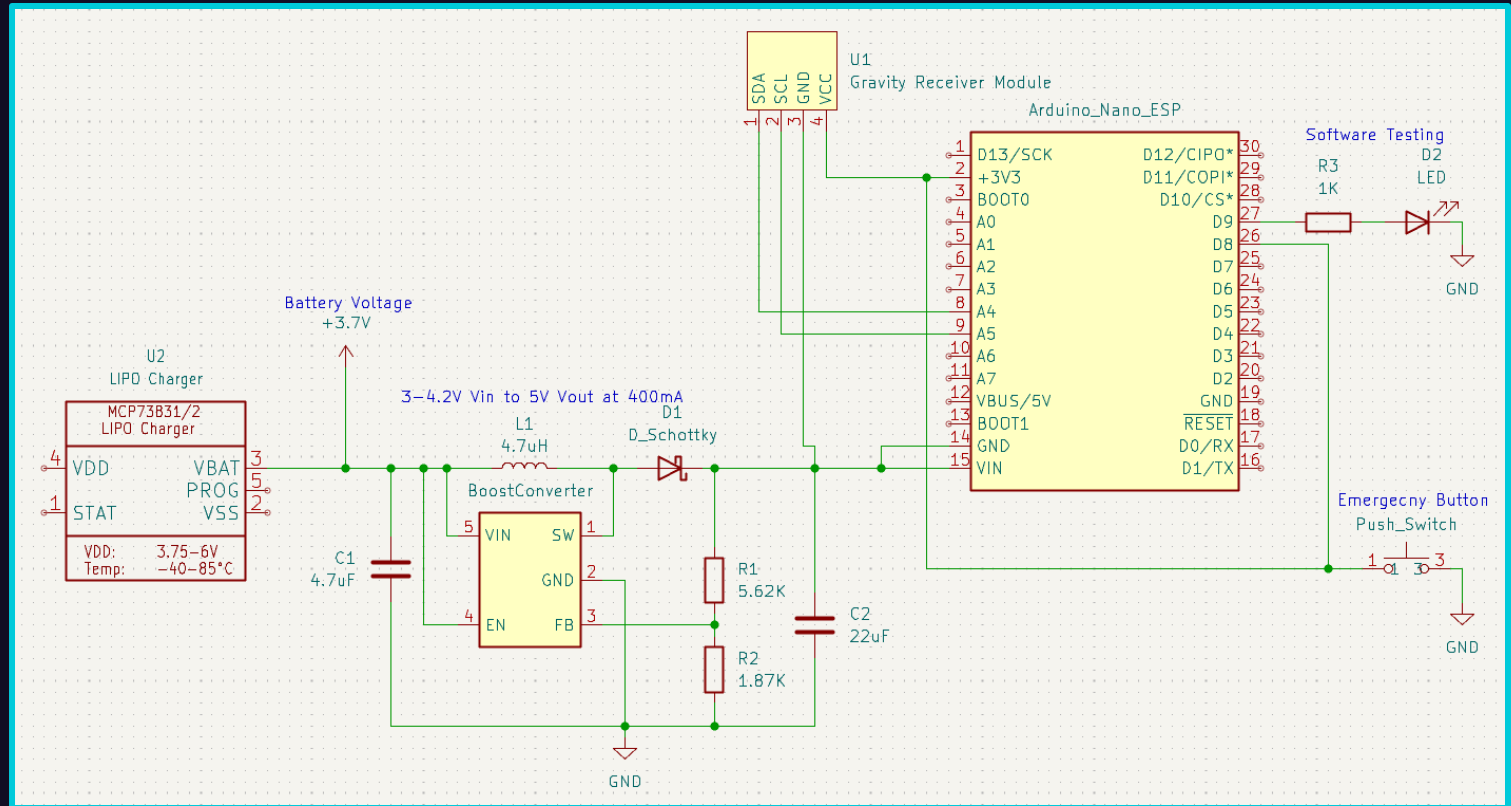


Microphone Operation & Performance

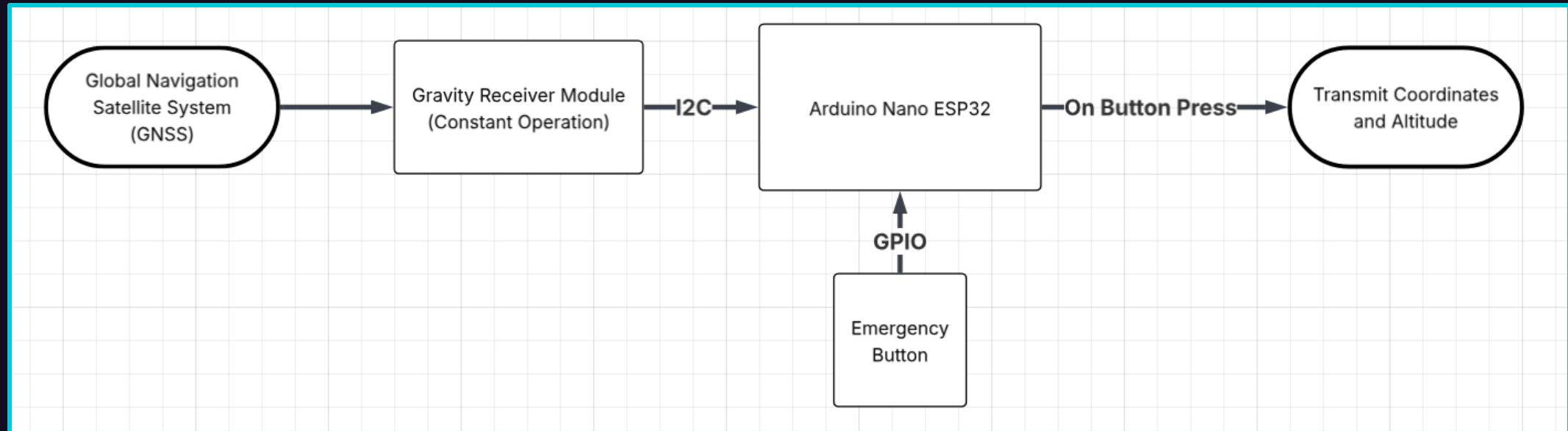


GPS Tracking Schematic

- MIC2288 Boost Converter
 - Layout/components obtained from datasheet
- Gravity Receiver Module
 - Built-in I2C pull-up resistors
 - A4-SDA and A5-SCL I2C pins for the Arduino
- LIPO Charger
 - Connected to power source and Boost Converter through VBAT to provide charging without taking apart the device



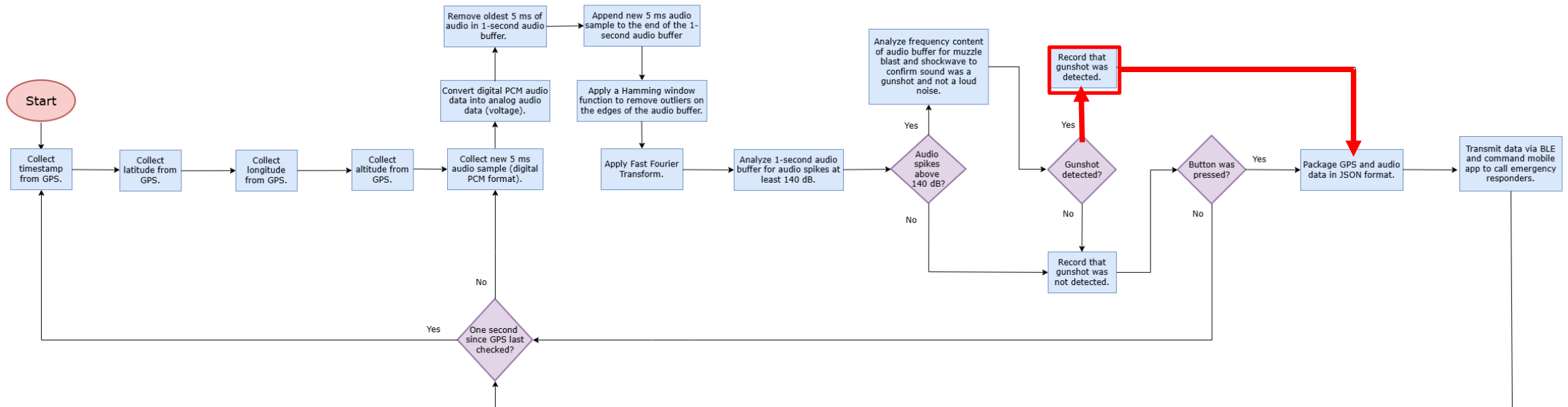
GPS Operation & Performance



► Software

Updated Block Diagram for Overall Design

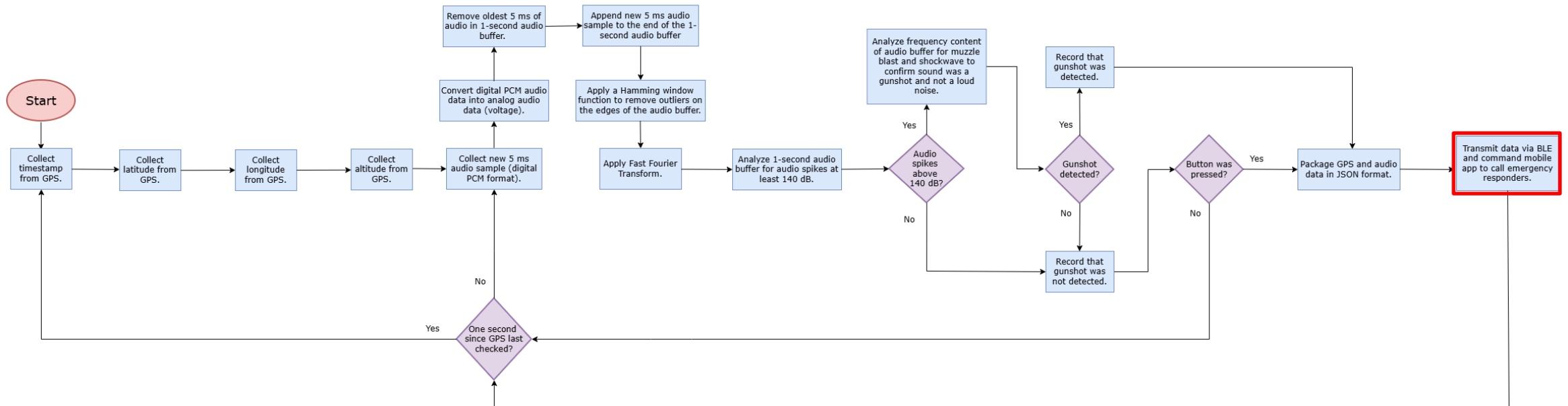
- Implemented changes and feedback discussed during PDR about overall software approach
- Change #1: Detecting a gunshot will automatically send data to mobile app, no button press needed
- Improvement #1: Concealability, user will not have to physically reach for button if gunshot is detected



► Software

Updated Block Diagram for Overall Design

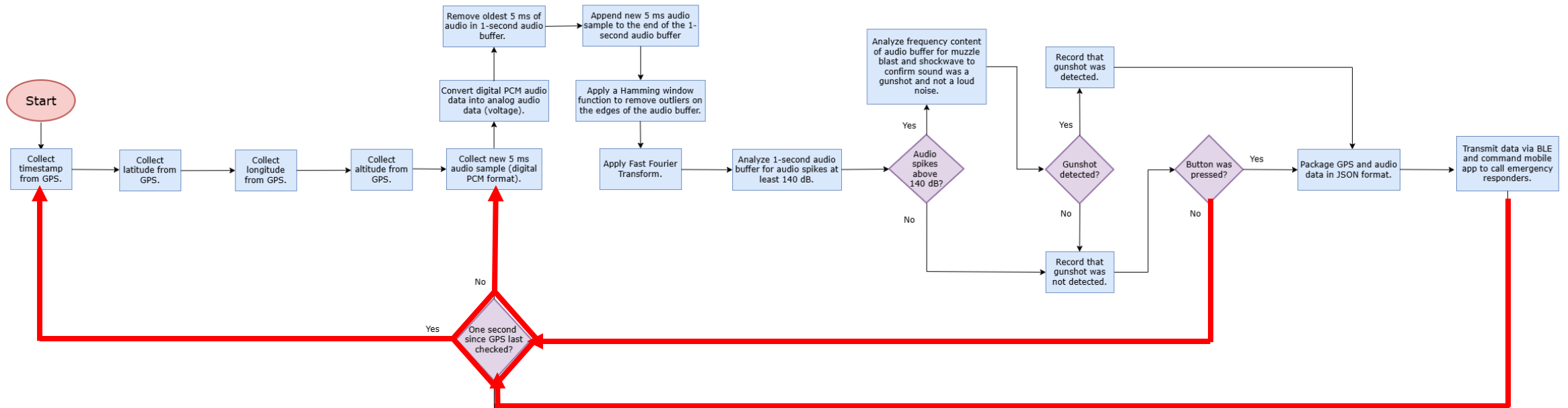
- Change #2: Indicated near the end of block diagram that WAD will command the mobile app to call emergency responders
- Improvement #2: Better clarification that mobile app will store emergency contact information, not WAD



Software

Updated Block Diagram for Overall Design

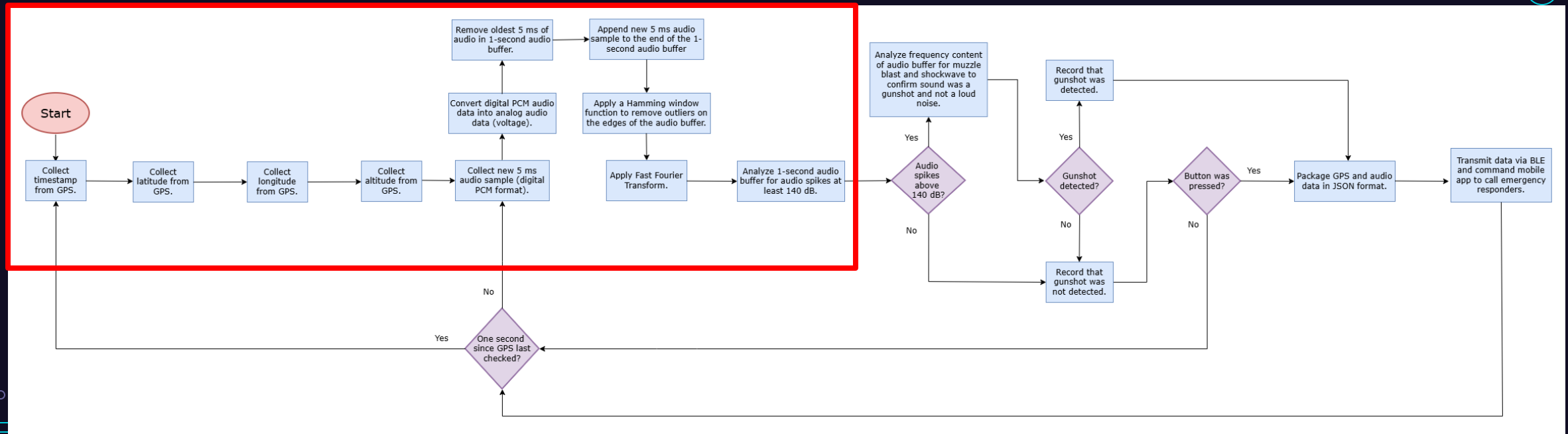
- Change #3: Added decision-making for collecting GPS data every one second
- Improvement #3: Allows for faster, real-time audio sampling since GPS data collection can be bypassed



► Software

Updated Block Diagram for Overall Design

- Change #4: Added more details on GPS data collection, audio collection and processing, and gunshot detection
- More discussion in later slides



► Software

Wireless Data Transmission

- Transmitting data to mobile app via Bluetooth Low-Energy (BLE)
- Data will be organized in JSON formatted packet

Data Packet Contents:

Information	Unit	Data Type in Code	Size (in bytes)
Timestamp (CST)	YYYY-MM-DD hh:mm:ss	C-type string	19
Latitude	Degrees	double	8
Longitude	Degrees	double	8
Altitude	Meters	double	8
Max Audio Reading	dB	int	4
If gunshot was detected	True/False	Boolean	1
If emergency responders need to be called	True/False	Boolean	1
		Total Size (in bytes):	49

► Software

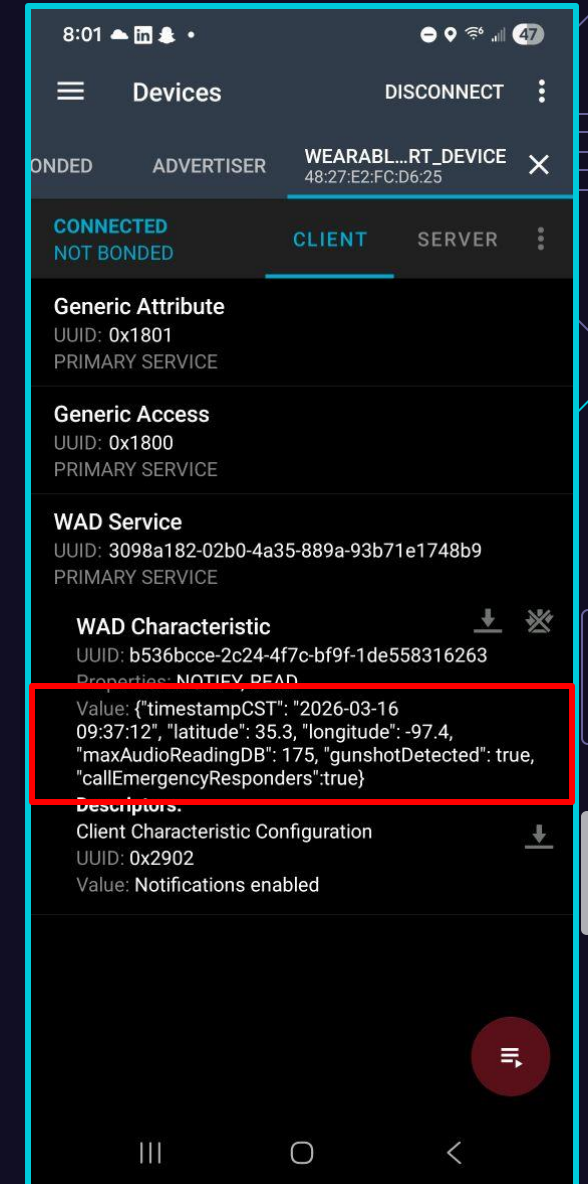
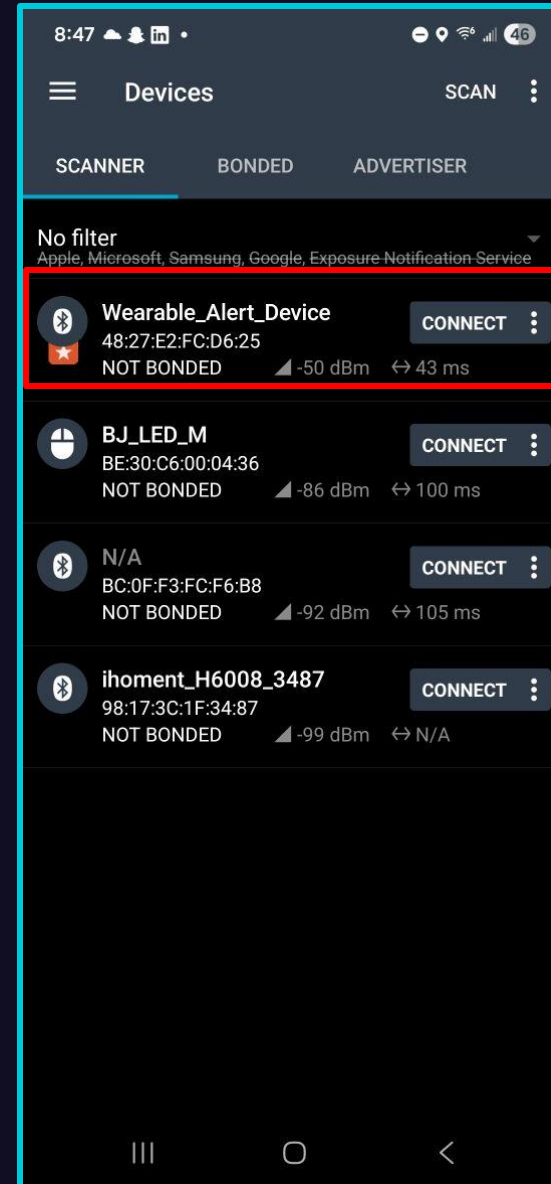
Wireless Data Transmission

Mobile App:

- To simulate mobile app, using a Bluetooth serial messaging app called nRF Connect for Mobile
- Available for Android and IOS, no account needed
- Can connect to the WAD via BLE using the app and receive/send messages to the device

Testing:

- If data from WAD is received on nRF Connect when the button is pressed or gunshot is detected, wireless data transmission is a success



► Software

Wireless Data Transmission

Important Note:

- For the mobile app to correctly connect to a WAD and not any other Bluetooth devices nearby, it needs to somehow implement the Service UUID¹ and Characteristic UUID¹
- Service UUID¹ and Characteristic UUID¹ are 128-bit codes that uniquely identify a Bluetooth device as a WAD

Service UUID: 3098A182-02B0-4A35-889A-93B71E1748B9

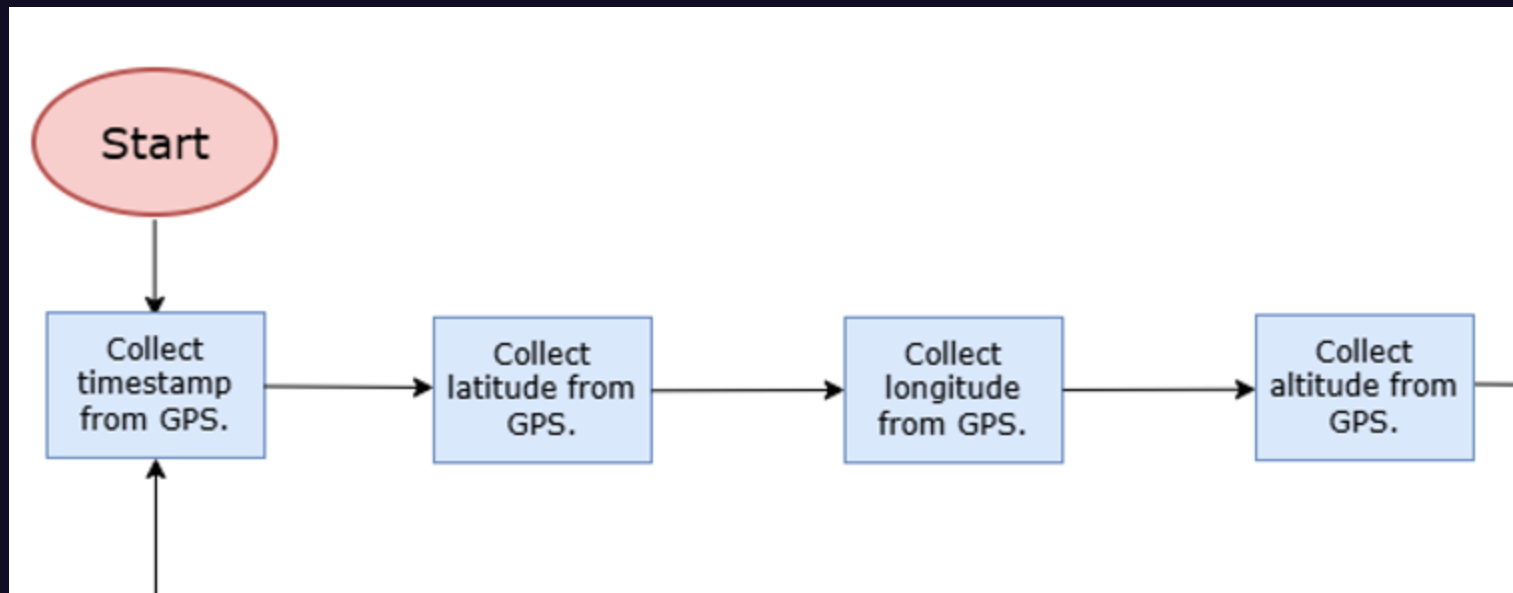
Characteristic UUID: B536BCCE-2C24-4F7C-BF9F-1DE558316263

¹UUID – Universally Unique Identifier

► Software

GPS Data Collection

- Using official software library developed for the Gravity Receiver Module chip



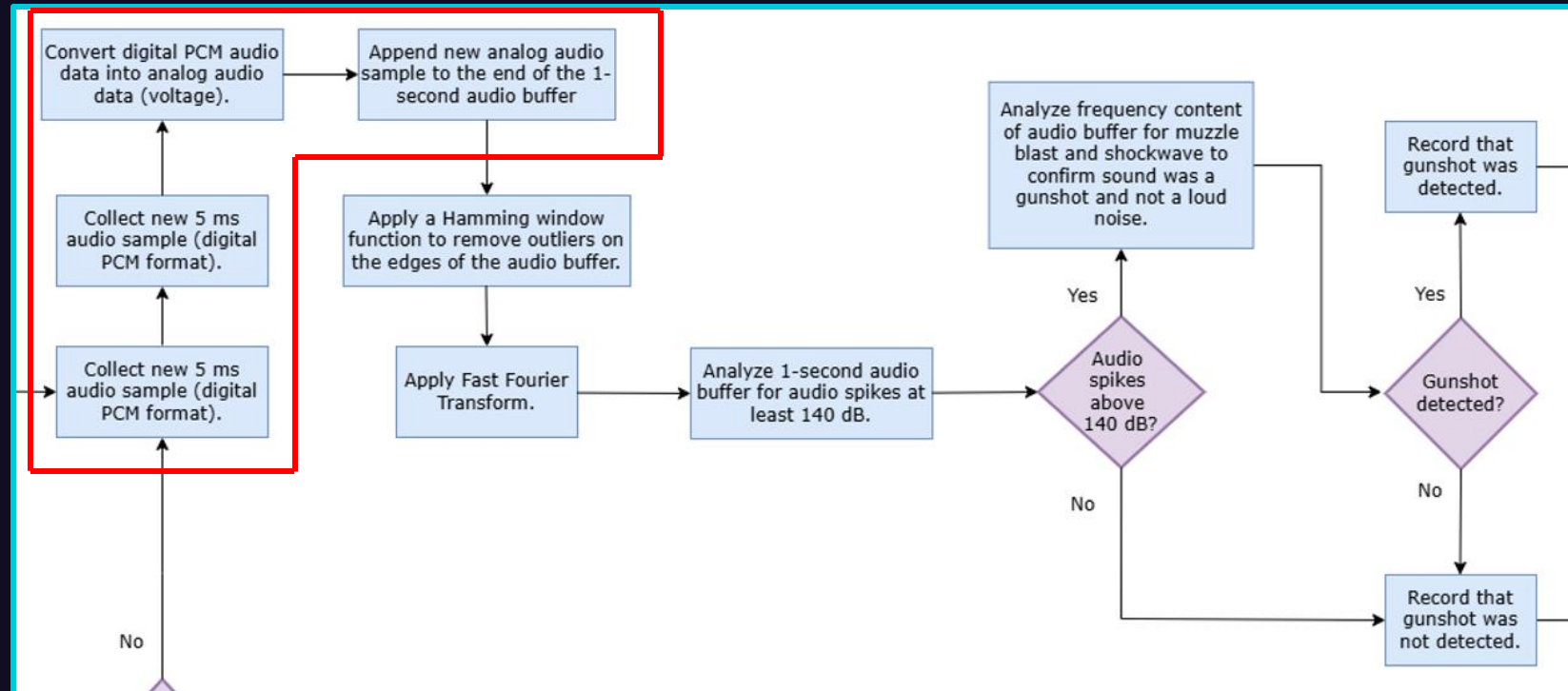
► Software

Gunshot Detection Diagram

- Updated gunshot detection process to occur continuously and in real time

Audio Data Collection:

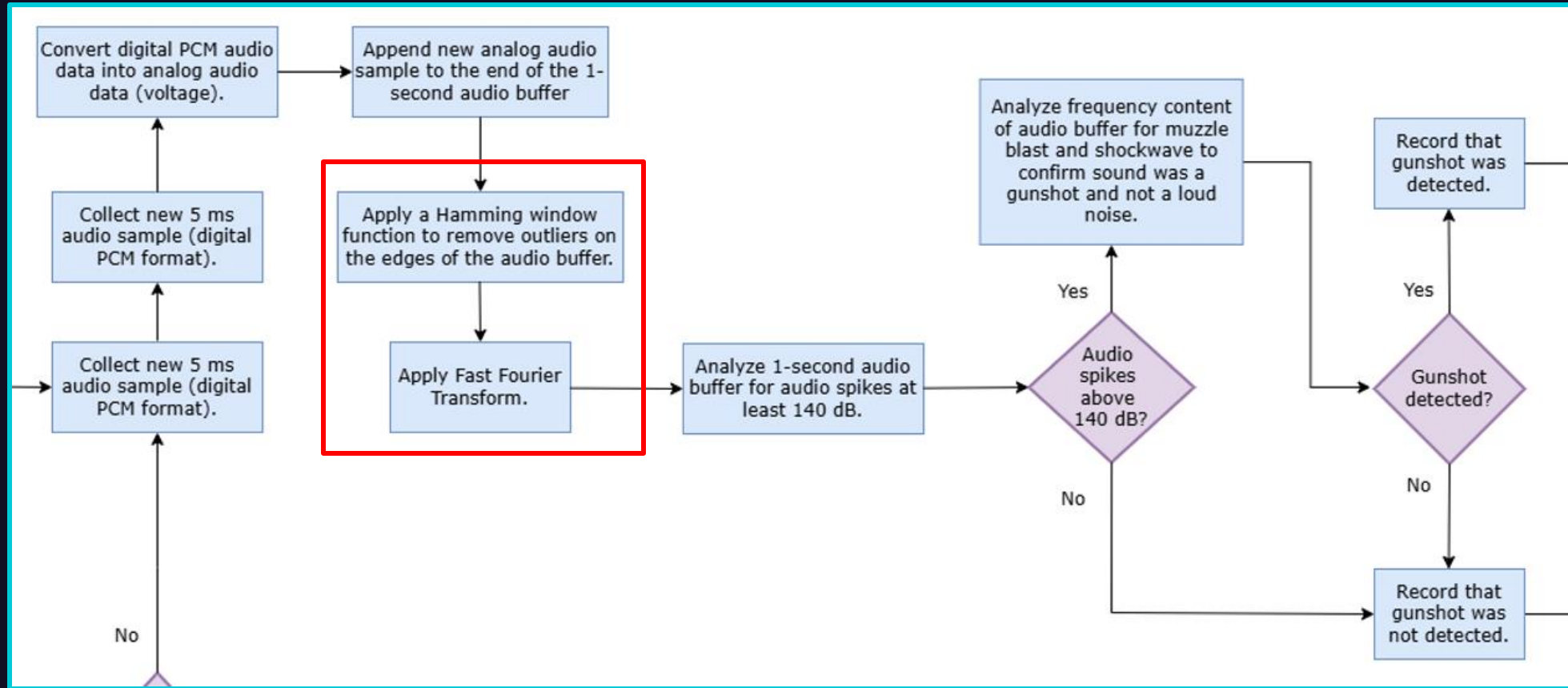
- A buffer stores the last second of audio
- Newest 5 ms of audio is continually fed into the buffer for real-time monitoring



► Software

Gunshot Detection Diagram

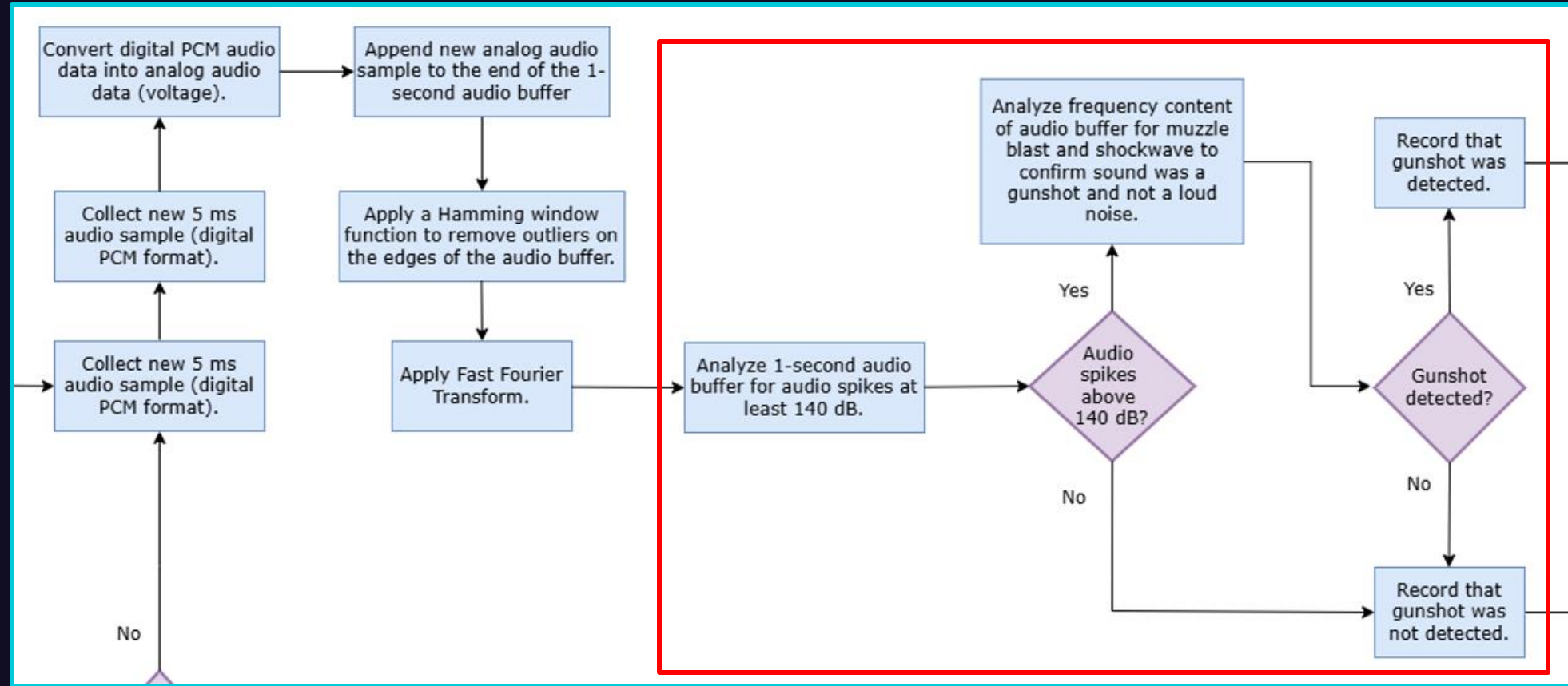
Audio Signal Processing:



► Software

Gunshot Detection Diagram

Gunshot Detection:



Test Plan

	Microphone	Position Module	Battery	Button/Bluetooth
What needs to be tested?	Max dB, Signal Processing, Fall Off dB	Elevation Accuracy, Latitude and Longitude Accuracy	Battery Life, Time to Charge	How Many Presses, Bluetooth Connectivity
How are we going to test it?	Actual field testing of software	Comparison to phone position and actual elevation position	Timing of mAh drain, Timing of Charge in real world	Button rating from datasheet, and actual device connection for bluetooth

► Revised Budget Plan

Item	Description	Quantity	Cost	Use
1	Arduino Nano ESP32	2	\$38.60	Main Micro controller for project
2	Lithium Ion Battery	2	\$39.90	Main Power supply for project
3	Micro-Lipo Charger	2	\$11.90	Rechargeable module for Battery Pack
4	Gravity: GNSS GPS	2	\$35.80	GPS Module for Project
5	I2S MEMS Microphone	2	\$9.80	Microphone Module for GS Detection
6	320E11RED	2	\$4.18	Button for panic device
7	MIC2288YD5-TR	2	\$1.46	Booster DC DC Converter
8	Breakout Board	2	\$2.50	Breakout board for Booster converter
9	SCHOTTKY RECTIFIER	3	\$0.87	Schottky diode for GPS Module



Questions?